Programming Fundamentals

Lab 4 Submission

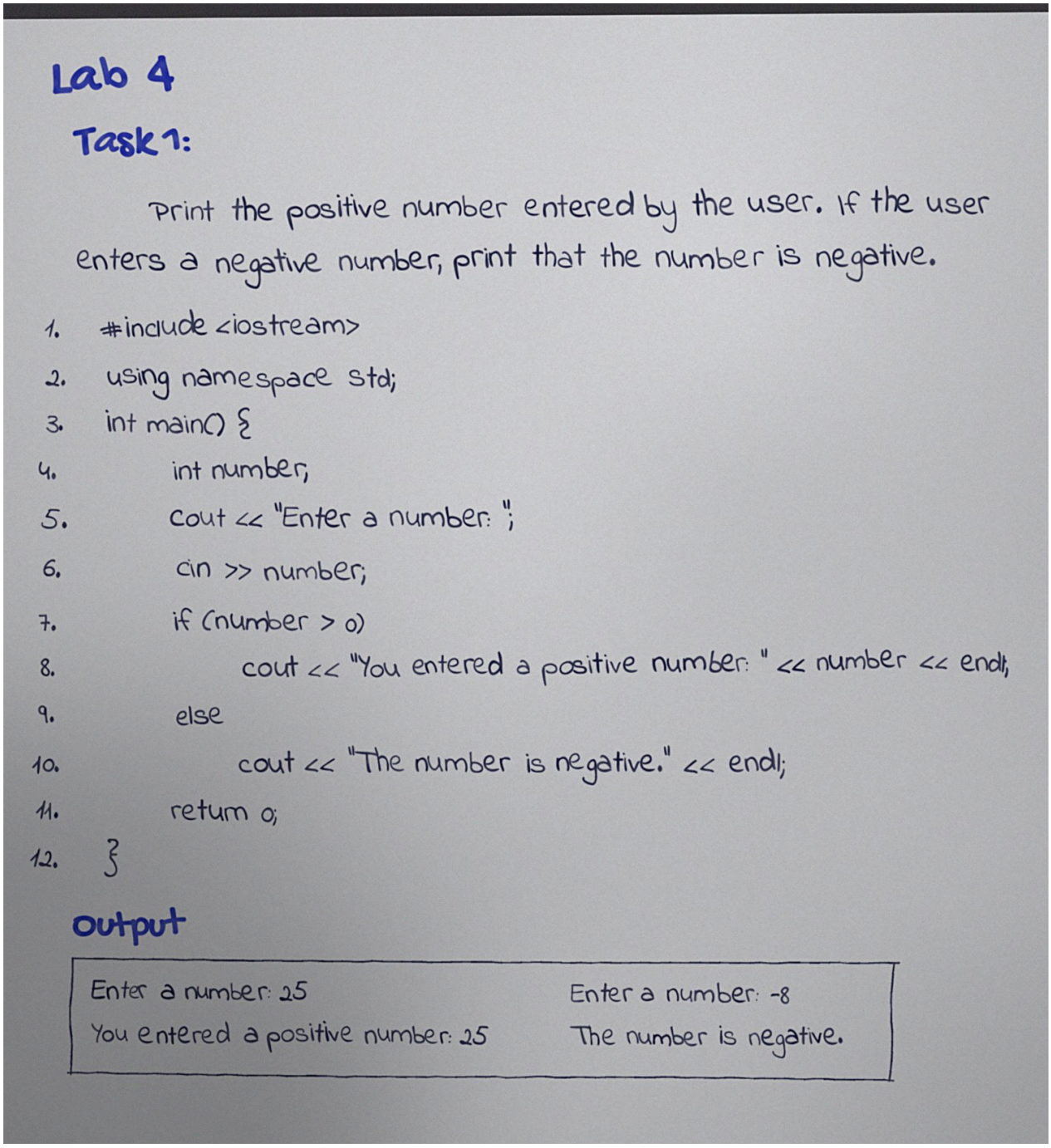
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Lab 4 – Task 1

Question

Write a program to print positive number entered by the user. If the user enters negative number print number entered is positive otherwise print number is negative.

Handwritten Code



Code Written on Phone / Desktop

```cpp
#include <iostream>
using namespace std;
int main() {
    int number;
    cout << "Enter a number: ";
    cin >> number;
    if (number > 0)
        cout << "You entered a positive number: " << number << endl;
    else
        cout << "The number is negative." << endl;
    return 0;
}
```

Output

```
Enter a number: 25
You entered a positive number: 25

Enter a number: -8
The number is negative.
```

Explanation

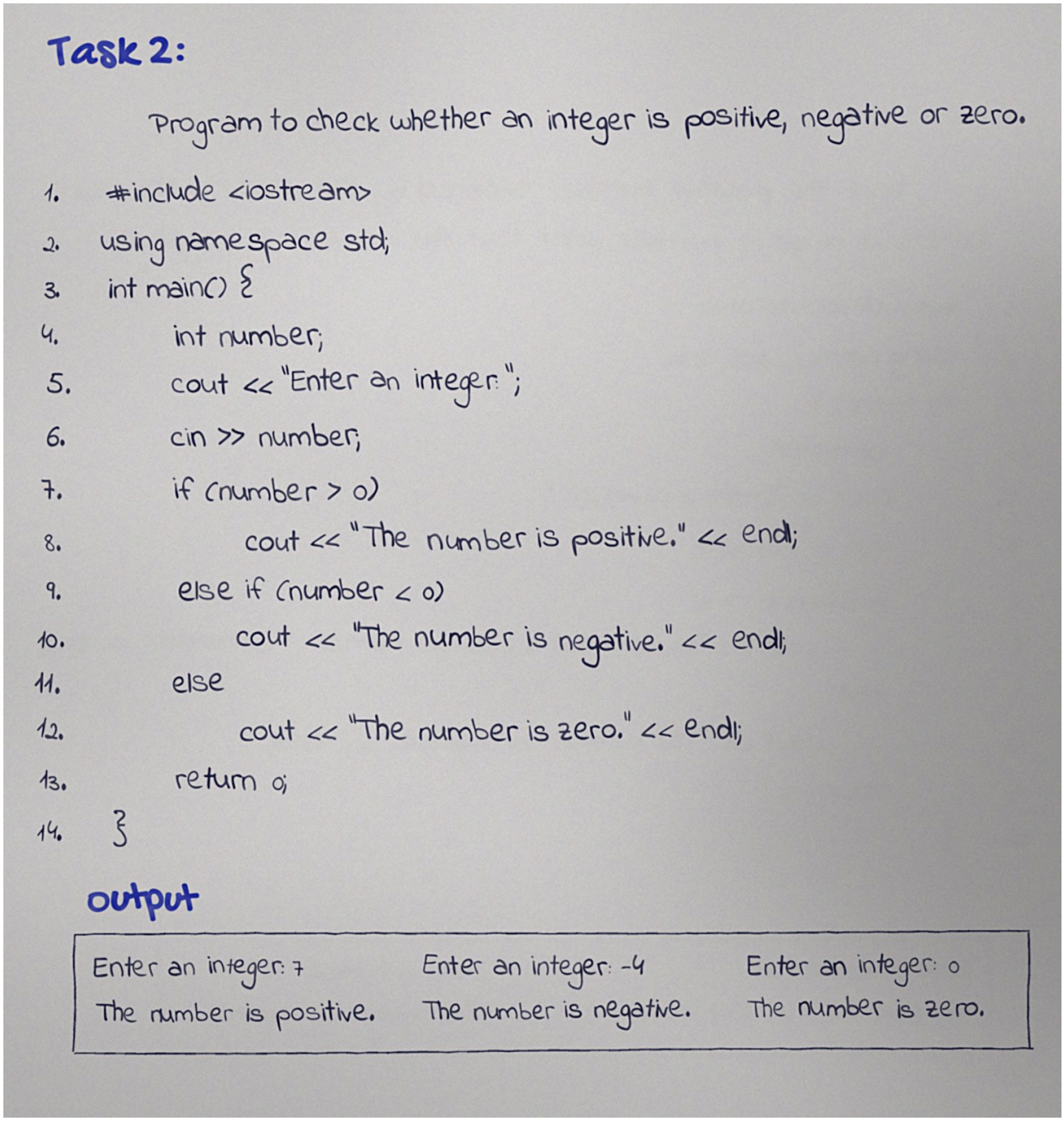
- The `if` checks the condition `number > 0`.
- If it is true, the number is positive and it is printed. If it is false, the `else` part runs and prints that the number is negative.
- The output shows two runs, one with 25 and one with -8.

Lab 4 – Task 2

Question

Program to check whether an integer is positive, negative or zero.

Handwritten Code



Code Written on Phone / Desktop

```cpp
#include <iostream>
using namespace std;
int main() {
    int number;
    cout << "Enter an integer: ";
    cin >> number;
    if (number > 0)
        cout << "The number is positive." << endl;
    else if (number < 0)
        cout << "The number is negative." << endl;
    else
        cout << "The number is zero." << endl;
    return 0;
}
```

Output

```
Enter an integer: 7
The number is positive.

Enter an integer: -4
The number is negative.

Enter an integer: 0
The number is zero.
```

Explanation

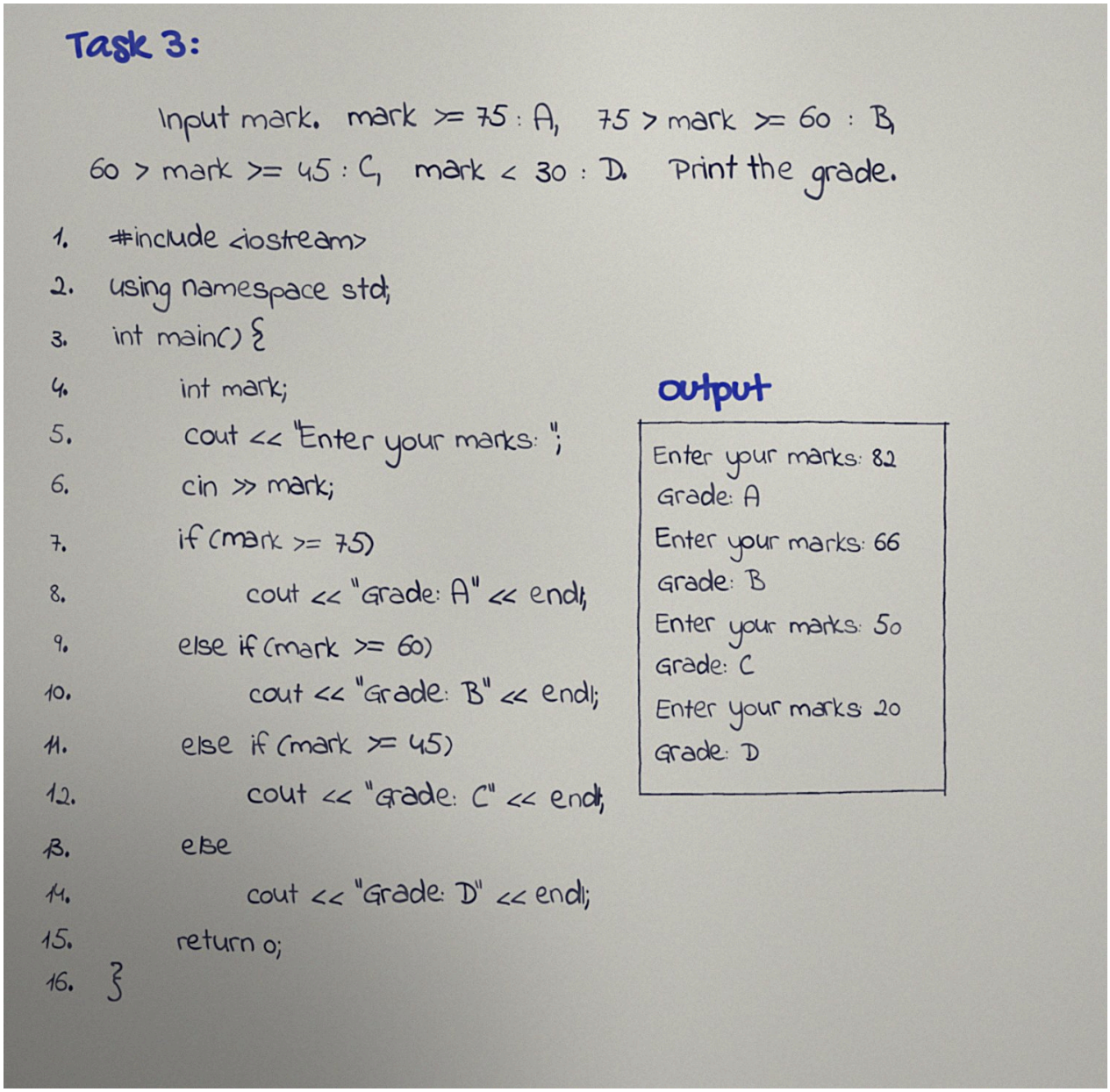
- This uses an `if ... else if ... else` ladder because there are three possible answers.
- First it checks `number > 0`, then `number < 0`. If both are false, the number can only be 0, so the last `else` prints zero.

Lab 4 – Task 3

Question

```
Input   : Mark
Process : If mark greater than and equal to 75, score will be A
          If mark less than 75 and greater than and equal to 60, score will be B
          If mark less than 60 and greater than and equal to 45, score will be C
          If mark less than 30, score will be D
Output  : Print the grade of your score.
```

Handwritten Code



Task 3:

Input mark. mark >= 75 : A, 75 > mark >= 60 : B,
60 > mark >= 45 : C, mark < 30 : D. Print the grade.

```
1.  #include <iostream>
2.  using namespace std;
3.  int main() {
4.      int mark;
5.      cout << "Enter your marks: ";
6.      cin >> mark;
7.      if (mark >= 75)
8.          cout << "Grade: A" << endl;
9.      else if (mark >= 60)
10.         cout << "Grade: B" << endl;
11.     else if (mark >= 45)
12.         cout << "Grade: C" << endl;
13.     else
14.         cout << "Grade: D" << endl;
15.     return 0;
16. }
```

output

```
Enter your marks: 82
Grade: A
Enter your marks: 66
Grade: B
Enter your marks: 50
Grade: C
Enter your marks: 20
Grade: D
```

Code Written on Phone / Desktop

```cpp
#include <iostream>
using namespace std;
int main() {
    int mark;
    cout << "Enter your marks: ";
    cin >> mark;
    if (mark >= 75)
        cout << "Grade: A" << endl;
    else if (mark >= 60)
        cout << "Grade: B" << endl;
    else if (mark >= 45)
        cout << "Grade: C" << endl;
    else
        cout << "Grade: D" << endl;
    return 0;
}
```

Output

```
Enter your marks: 82
Grade: A

Enter your marks: 66
Grade: B

Enter your marks: 50
Grade: C

Enter your marks: 20
Grade: D
```

Explanation

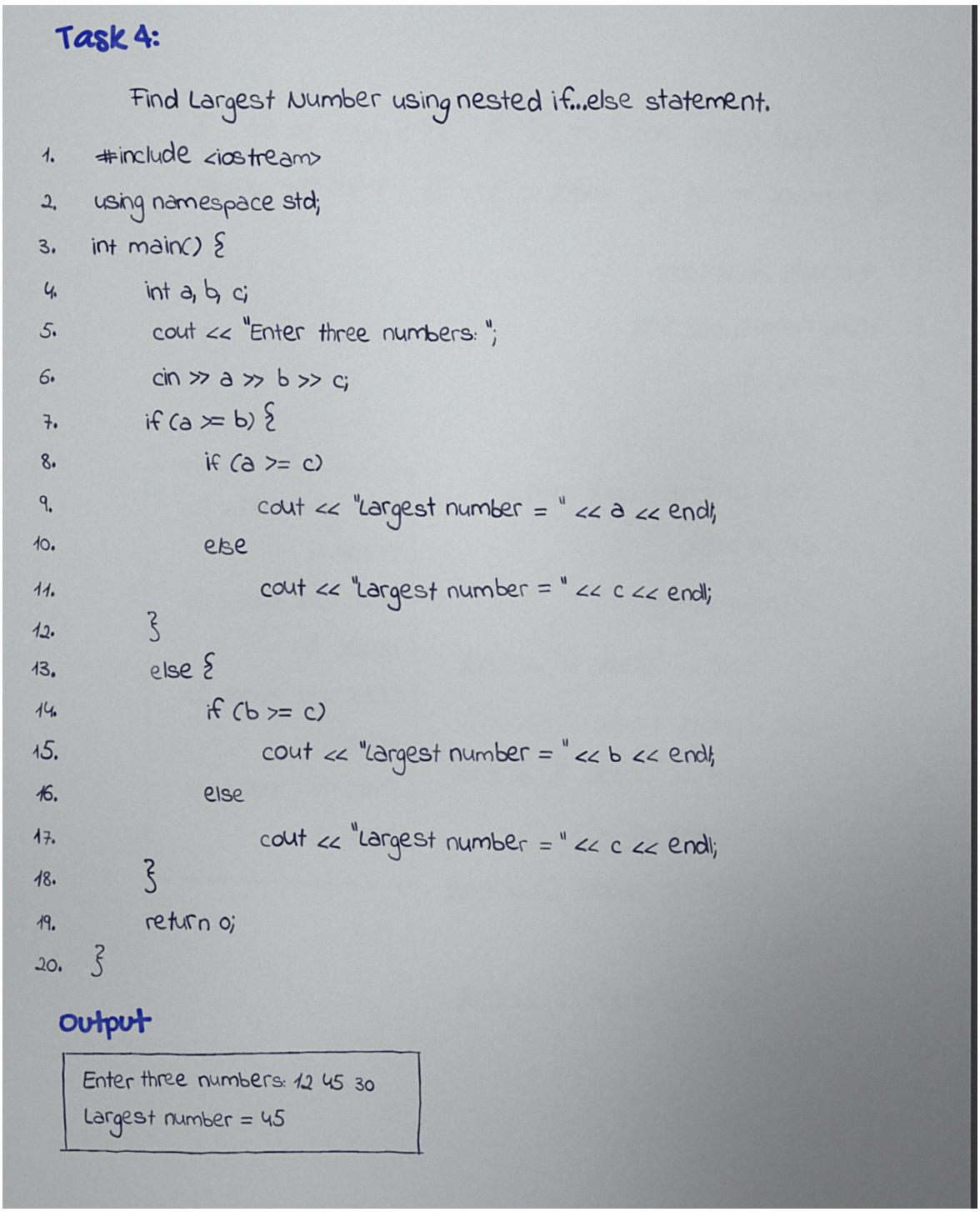
- The conditions are checked from the highest grade down, so each `else if` only needs to check the lower limit.
- For example, if the mark reaches `else if (mark >= 60)` it is already less than 75, so it gets a B.
- The question does not give a grade for marks from 30 to 44, so every mark below 45 gets D in the final `else`.

Lab 4 – Task 4

Question

Find Largest Number Using Nested if...else statement.
Check whether the number entered by the user is positive or not. If it is positive then calculate how many digits the number have.

Handwritten Code (Largest number)



Task 4:

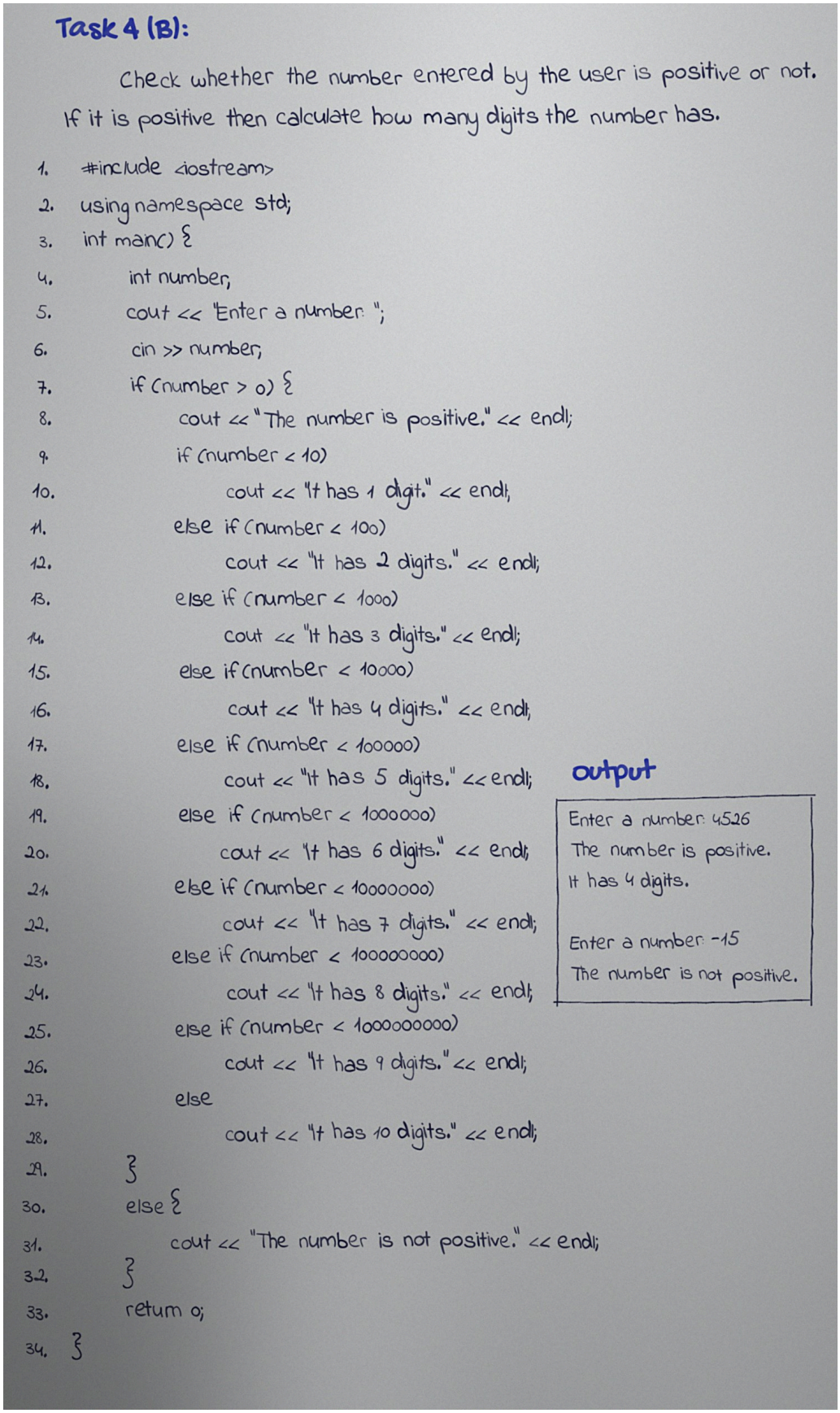
Find Largest Number using nested if...else statement.

```
1.  #include <iostream>
2.  using namespace std;
3.  int main() {
4.      int a, b, c;
5.      cout << "Enter three numbers: ";
6.      cin >> a >> b >> c;
7.      if (a >= b) {
8.          if (a >= c)
9.              cout << "Largest number = " << a << endl;
10.         else
11.             cout << "Largest number = " << c << endl;
12.     }
13.     else {
14.         if (b >= c)
15.             cout << "Largest number = " << b << endl;
16.         else
17.             cout << "Largest number = " << c << endl;
18.     }
19.     return 0;
20. }
```

Output

```
Enter three numbers: 12 45 30
Largest number = 45
```

Handwritten Code (Positive number and its digits)



Task 4 (B):

Check whether the number entered by the user is positive or not. If it is positive then calculate how many digits the number has.

```
#include <iostream>
using namespace std;
int main() {
    int number;
    cout << "Enter a number: ";
    cin >> number;
    if (number > 0) {
        cout << "The number is positive." << endl;
        if (number < 10)
            cout << "It has 1 digit." << endl;
        else if (number < 100)
            cout << "It has 2 digits." << endl;
        else if (number < 1000)
            cout << "It has 3 digits." << endl;
        else if (number < 10000)
            cout << "It has 4 digits." << endl;
        else if (number < 100000)
            cout << "It has 5 digits." << endl;
        else if (number < 1000000)
            cout << "It has 6 digits." << endl;
        else if (number < 10000000)
            cout << "It has 7 digits." << endl;
        else if (number < 100000000)
            cout << "It has 8 digits." << endl;
        else if (number < 1000000000)
            cout << "It has 9 digits." << endl;
        else
            cout << "It has 10 digits." << endl;
    }
    else {
        cout << "The number is not positive." << endl;
    }
    return 0;
}
```

output

```
Enter a number: 4526
The number is positive.
It has 4 digits.

Enter a number: -15
The number is not positive.
```

Code Written on Phone / Desktop (Largest number)

```cpp
#include <iostream>
using namespace std;
int main() {
    int a, b, c;
    cout << "Enter three numbers: ";
    cin >> a >> b >> c;
    if (a >= b) {
        if (a >= c)
            cout << "Largest number = " << a << endl;
        else
            cout << "Largest number = " << c << endl;
    }
    else {
        if (b >= c)
            cout << "Largest number = " << b << endl;
        else
            cout << "Largest number = " << c << endl;
    }
    return 0;
}
```

Output (Largest number)

```
Enter three numbers: 12 45 30
Largest number = 45
```

Code Written on Phone / Desktop (Positive number and its digits)

```cpp
#include <iostream>
using namespace std;
int main() {
    int number;
    cout << "Enter a number: ";
    cin >> number;
    if (number > 0) {
        cout << "The number is positive." << endl;
        if (number < 10)
            cout << "It has 1 digit." << endl;
        else if (number < 100)
            cout << "It has 2 digits." << endl;
        else if (number < 1000)
            cout << "It has 3 digits." << endl;
        else if (number < 10000)
            cout << "It has 4 digits." << endl;
        else if (number < 100000)
            cout << "It has 5 digits." << endl;
        else if (number < 1000000)
            cout << "It has 6 digits." << endl;
        else if (number < 10000000)
            cout << "It has 7 digits." << endl;
        else if (number < 100000000)
            cout << "It has 8 digits." << endl;
        else if (number < 1000000000)
            cout << "It has 9 digits." << endl;
        else
            cout << "It has 10 digits." << endl;
    }
    else {
        cout << "The number is not positive." << endl;
    }
    return 0;
}
```

Output (Positive number and its digits)

```
Enter a number: 4526
The number is positive.
It has 4 digits.

Enter a number: -15
The number is not positive.
```

Explanation

- Largest number: the outer `if` compares a and b. Inside it, the bigger of those two is compared with c, so the largest of all three is printed.
- Digits: the outer `if` checks that the number is positive. The inner `if ... else if` checks the size: less than 10 means 1 digit, less than 100 means 2 digits, and so on.
- If the number is not positive, the outer `else` runs and no digits are counted.

END OF ASSIGNMENT

```
return 0;   // program finished successfully
```

Thank you for reading!